

INFORMATICS INSTITUTE OF TECHNOLOGY

Foundation Certificate in Higher Education

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II. Abstract

An overview of information systems (IS), which are crucial elements of contemporary companies, is given in this study abstract. Hardware, software, data, and people all come together in information systems, which are used to manage, process, and convey information. The relevance of data management in information systems is then examined in the paper. Each organization's success depends on its ability to handle data effectively, which enables the fast and correct processing of information.

The importance of information systems in supporting company operations including supply chain management, customer relationship management, human resource management and much more are also highlighted in the paper. These procedures may be made more efficient and effective by using information systems.

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1. Introduction

1.1. Introduction to the organization

- Seoul, South Korea serves as the corporate headquarters for the multinational conglomerate Samsung. The business was established in 1938 and has since grown to rank among the biggest in the world for electronics. Mobile phones, televisions, household appliances, and computer hardware are just a few of the many goods that Samsung makes.

- One of the most notable divisions of Samsung is its mobile business. The company's business creates a range of Android-powered smartphones and tablets. Samsung is renowned for its excellent displays and cameras, and its flagship smartphones are frequently among the most well-liked products available.



Figure 1: 1.1 Samsung Logo

- The mobile division is also in charge of creating Samsung's mobile software, including its unique One UI user interface. The organization also collaborates closely with carriers and merchants to distribute Samsung products around the world.

- Despite competition from other smartphone producers like Apple and Huawei, Samsung is still a significant participant in the mobile market. The company has been able to hold onto its place in the market because of its capacity to produce a wide range of products at various pricing ranges.

1.2. Introduction to the organization’s usage of information systems in activities

- Information systems are essential to Samsung Mobile's ability to serve consumers, optimize processes, and boost productivity. In supply chain management, these methods are used to procure and ship items quickly.

- Engineers can use information systems to test and simulate design concepts during design and development. Information systems are used in manufacturing facilities to regulate quality, inventory, and production.

- Samsung also uses information systems to analyze consumer behavior, identify market trends, and develop niche marketing strategies.

- Via information systems, sales channels are maintained, customer orders are tracked, and payments are processed.

2. Information system types and their examples

2.1. Management support



Figure 2: 2.1.1 Knowledge Management System

The management of Samsung's information systems is committed to utilizing technology to improve productivity and efficiency through growth-oriented strategies that are in line with company objectives. Samsung remains at the forefront of the technology sector because of their strong management support System.

Some examples of MSIS used by Samsung and their functions are:

- **Knowledge Management Systems:** Knowledge can be captured, stored, and shared within an organization with the assistance of these systems, which can help boost productivity, creativity, and teamwork among employees.
- **Project Management Systems:** These systems aid in the planning, scheduling, resource allocation, and risk management of projects from start to finish.
- **Marketing Automation Systems:** In order to increase marketing efficiency and effectiveness, these technologies assist in automating operations including lead generation, email marketing, social media management, and customer segmentation.
- **Financial Management Systems:** Billing, accounting, budgeting, and financial reporting are just a few of the financial transactions that these systems assist with managing, which can aid with financial control and decision-making.

These are but a few types of the numerous management support systems that are available. A corporation like Samsung's particular operating systems would depend on its own requirements and priorities.



Figure 3: 2.1.2 Financial Management System

2.2. Support of knowledge work

Samsung is a sizable international corporation that supports its operations with a range of knowledge work systems (KWS).

Such instances are as follows:

- **Collaboration Tools:** Samsung encourages communication and teamwork among its employees by utilizing a variety of collaboration technologies, including video conferencing, instant messaging, and virtual whiteboards. As a result, they are able to collaborate better, exchange ideas and information, and make choices right away.
- **Enterprise Resource Planning Systems:** Samsung employs ERP systems to control their inventories, supply chain, and other corporate operations. These systems deliver real-time information, allowing staff to take decisions based on data and act fast in response to market developments.
- **Customer Relationship Management Systems:** Samsung manages its customer relationships through CRM systems. They may handle customer care requests, track client preferences and behaviors, and offer consumers individualized support thanks to these tools.



Figure 4: 2.2.1 Support Of Knowledge Work

- **Data Analytics:** Samsung analyzes market trends and consumer data using data analytics techniques. They may use this information to recognize new trends, predict client requirements, and create new goods and services.

Overall, KWS play a crucial part in assisting Samsung's operations, assisting staff members in working more productively, making wiser judgments, and ultimately fostering corporate success.

2.3. Operational support system

Software tools used by Samsung's operational support system (OSS) to handle network infrastructure, customer service, and billing. The OSS enables real-time network performance monitoring, sophisticated customer assistance, and invoicing and revenue management. The OSS is essential to Samsung's success in the cutthroat technology sector.

Some of the key features of Samsung's OSS include:

- **Service management:** Mobile networks, internet, and Internet of Things services are just a few of the services that Samsung's OSS aids in managing. The technology can monitor service usage and spot possible issues immediately.
- **Network management:** In monitoring network performance, spotting possible network problems, and optimizing network resources, Samsung's OSS aids in managing its network infrastructure.
- **Customer support:** Customer support technologies like call center administration, client feedback management, and self-service portals are all included in Samsung's open source software (OSS).
- **Billing and revenue management:** Billing systems, revenue assurance measures, and fraud control are just a few of the instruments for managing revenue and billing that are part of Samsung's OSS.

In general, Samsung's OSS aids the business in managing operations more effectively and efficiently, ensuring that services are reliably delivered and consumers are happy.



Figure 5: 2.3.1 Operational Support System

3. Components used in Management information system.

3.1. Hardware used in Samsung company.

Management is able to make choices with the use of an information system known as a management support information system (MSIS). As a company, Samsung also uses hardware in its MSIS. Nonetheless, the following hardware elements are frequently seen in an MSS:

- **Personal Computers:** Computers are frequently used in MSSs to provide managers access to the tools and programs they need to make decisions.
- **Servers:** Servers are used to store the information and software that managers need to make decisions.
- **Storage Devices**: Data created by the system is stored on storage devices such hard drives and network-attached storage.
- **Networking Equipment**: The MSS is connected to other systems and devices through networking hardware like switches and routers.

3.2. Software used in Samsung company.

Depending on the particular requirements of the organization, a wide variety of software types can be applied in the management support information system of the Samsung Company.

- **Project Management software:** To plan, oversee, and manage activities, timelines, funds, and resources in projects, project management software is employed.
- **Business Intelligence (BI) software:** BI software is used to gather, examine, and present data from a variety of sources to assist management in making better decisions.
- **Customer Relationship Management software:** By monitoring behavior and preferences and automating sales and marketing procedures, CRM software assists firms in managing client interactions and relationships.

3.3. Data used in Samsung Company

Management Support Systems (MSS) are created to offer timely, accurate, and relevant information to decision-makers to aid in their decision-making processes. The following data formats are frequently used in MSS:

- **Financial Data:** This data helps decision-makers to evaluate the financial performance of the organization and make informed financial decisions.

- **Sales and Marketing Data:** This data helps decision-makers to develop marketing strategies and make informed decisions about product development.
- **Operational Data:** This data helps decision-makers to optimize operational efficiency and make informed decisions about resource allocation.

3.4. People in Samsung Company

- **Management:** The top-level management of Samsung would be responsible for overseeing the company's MSS and making decisions based on the information provided by the system.
- **Data Analysts:** Data analysts would be responsible for collecting and analyzing data from various sources to provide insights and recommendations to decision-makers.
- **Operations Staff:** Operations staff would use the information provided by the MSS to optimize operational processes and make informed decisions about resource allocation.

3.5. Process in Samsung Company

The procedures involved in Samsung's MSS would probably be intricate and complicated, including several organizational divisions and functions. In general, a company's MSS may entail the following processes:

- **Data Collection:** The first step in an MSS is to collect data from various sources, both internal and external. Samsung would likely use data from sales and marketing, finance, production, research and development, and other sources to inform decision-making.
- **Data Storage and Management:** Once data is collected, it must be stored and managed in a way that is easily accessible and usable by decision-makers. Samsung would likely use databases, data warehouses, and other tools to store and manage their MSS data.
- **Data Analysis:** Data analysts would then use various tools and techniques to analyze the data, looking for patterns, trends, and insights that can inform decision-making.

Overall, the goal of the MSS is to provide decision-makers with accurate, relevant, and timely information to support their decision-making processes and ensure the long-term success of the company.

4. Security threats

4.1. What are the Security Threats

Insiders are just as capable of causing harm as external attackers, whether through the distribution of malware, spyware, or viruses. Information systems face a variety of security risks, such as:

4.1.1. Virus

Malicious code called a computer virus may destroy data, spam contacts, steal passwords, record keystrokes, and even seize control of the infected device. Someone must intentionally or unintentionally distribute the sickness for them to become infected.

4.1.2. Spyware

Even if a piece of software is downloaded for good intentions, it is still considered spyware if it is placed on a device without the end user's consent. Spyware includes things like Trojans, keystroke loggers, and adware.



Figure 6: 4.1.1 Spyware

4.1.3. Malware

Malicious software, or malware, has the ability to steal, encrypt, or erase personal data, track online behavior without the user's knowledge, or change a device's basic computing capabilities. Malware includes things like viruses and spyware.

4.1.4. Phishing attacks

Cybercriminals can deceive people into disclosing personal information, including usernames, passwords, and credit card numbers, through phishing attacks. Phishing attempts sometimes take the shape of phony emails, instant chats, or posts on social media that impersonate a trustworthy source, such a bank or online merchant.



Figure 7: 4.1.2 Phishing Attack

4.2. How to stop threats to insider security

4.2.1 Security policy first

Ensure that your policy outlines the restrictions on the use and disclosure of personal information regarding your employees, temporary workers, and other people who could be the subject of inquiries. The misuse of sensitive data may result in serious repercussions, including legal action. Indicate who has access to what information, when they have access to it, and who they may share it with.

4.2.2. Don't neglect physical security!

Card thieves can be stopped by adding two-factor authentication to keycards, such as using a PIN and a keycard, but compliant employees will still lend their cards and PINs to coworkers. From unlocked hard copies, thieves or too curious coworkers will steal vital material. Ensure that every employee has a lockable drawer in their desk or filing cabinet to protect confidential information.

4.2.3. Screen new hires

Such relationships can be found using services like Systems Research & Development's NORA (Non-Obvious Relationship Awareness). NORA is able to conduct personnel checks on current workers, potential hires, and subcontractors and vendors by merging data from seemingly unconnected company systems.

4.2.4. Use strong authentication!

Passwords are obsolete. Stronger passwords cause forests of post-it notes to appear on monitors, and password-cracking technology is fairly sophisticated. And a lot of workers exchange passwords.

The alternatives are pricey, and most firms cannot afford a widespread implementation. Using powerful multifactor authentication solely to more critical applications or systems, such as HR or accounting, is a more economical compromise.

5. Conclusion

In conclusion, information systems are a crucial part of contemporary company operations, and businesses must carefully examine how to utilize and manage them in order to maximize their potential advantages while reducing risks and difficulties. Businesses may increase productivity, enhance decision-making skills, and promote a culture of cooperation and creativity by utilizing the potential of information technology. In order to develop and manage information systems effectively for their respective companies, a company must provide serious consideration.

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7. Appendices

7.1 Workload Matrix

GIVEN TOPICS		Student ID					
		20222250	20222167	20222395	20222007	20222420	20222431
INTRODUCTION	Introduction To the Organization	0	25%	75%	0	0	0
	Usage Of Information Systems	0	25%	75%	0	0	0
I/S TYPES	Management support	0	0	0	0	0	100%
	Support of knowledge work	0	0	0	0	50%	50%
	Operational support system	0	0	0	0	100%	0
COMPONENTS USED IN MSIS	Hardware	25%	25%	0	50%	0	0
	Software	25%	25%	0	50%	0	0
	Data	25%	25%	0	50%	0	0
	People	50%	25%	0	25%	0	0
	Process	50%	25%	0	25%	0	0
SECURITY THREATS & SOLUTIONS		0	75%	25%	0	0	0
PRESENTATION DESIGNING AND CONTENT		10%	50%	10%	10%	10%	10%

Table 1: 7.1 Workload Matrix



Figure 8: 7.1 Workload Matrix

7.2 Meeting Agenda

Meeting No.	Venue	Duration	Date	Purpose
01.	Meeting at IIT, Via WhatsApp Group call	45 **min**	04-03-2023	Adding the members of the group to a WhatsApp group and appointing a group leader
02.	Via WhatsApp Group call, Teams	35 – 55 **min**	05-03-2023	Choosing a topic for the report and sharing the work among group members
03.	Via WhatsApp Group call, Teams	1**h** 45**min**	07-03-2023	Solving problems and correcting errors
04.	Via WhatsApp Group call	30 **min**	08-03-2023	Talk about gathering information and give a deadline.
05.	Meeting at IIT, Teams	50 **min**	10-03-2023	Completing the report and correcting its errors.
06.	Via WhatsApp Group call, Teams	2**h** 15 **min**	11-03-2023	Taking the main points of the report and making the presentation
07.	Via WhatsApp Group call, Teams	1**h** 30 **min**	12-03-2023	Solving problems and correcting errors

Table 2: 7.2 Meeting Agenda



Figure 9: 7.2 Meeting Agenda